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Abstract

This paper studies how real childcare fees in Australia evolve around two major subsidy reforms: the Higher Child Care Subsidy (HCCS) and the Cheaper Child Care Bill (CCCB). The Australian childcare sector is characterised by heavily subsidised demand, geographically segmented markets, and relatively inelastic short-run supply due to regulatory and capacity constraints, implying that policy changes may be reflected in observed fee dynamics.

Using a quarterly panel of SA3 regions across Australian capital cities from 2018 to 2024, the analysis estimates a panel interrupted time series model with unit fixed effects and flexible time trends, allowing for level and trend changes in real hourly fees following each reform.

Results indicate that observed fees increase following the HCCS, with both level shifts and higher post-policy growth, while effects associated with the CCCB are smaller and less persistent. Robustness checks yield similar patterns. Placebo tests reveal strong underlying time dependence in fees, suggesting results should be interpreted as policy-associated deviations from trend rather than strictly causal effects.

1 Introduction

Early childhood education and care in Australia is a highly subsidised, supply-constrained service sector in which observed prices reflect the interaction of provider pricing decisions, regulatory requirements, and government subsidy design. Childcare services are delivered through a mix of private and not-for-profit providers operating in geographically segmented markets, where capacity constraints, staffing ratios, and compliance costs limit short-run supply responsiveness. As a result, changes in policy settings that affect childcare demand or subsidy generosity may be reflected in both utilisation patterns and observed fee dynamics.

The Australian Child Care Subsidy (CCS) system provides means-tested assistance by reimbursing a proportion of childcare fees directly to providers on behalf of eligible households. Families face a gross fee set by providers, with their effective out-of-pocket cost determined by subsidy rates that vary with income, activity, and family composition. This structure implies that policy reforms can affect both the marginal price of childcare and the distribution of subsidised demand across households, with potential implications for observed fee setting in equilibrium.

This paper examines two major reforms to the subsidy system: the Higher Child Care Subsidy (HCCS) and the Cheaper Child Care Bill (CCCB). The HCCS increased subsidy generosity within the existing framework, including more progressive rates and enhanced support for families with multiple children in care, thereby lowering the marginal cost of additional childcare use for a substantial subset of households. In contrast, the CCCB primarily expanded eligibility and increased subsidy coverage for higher-income households, broadening access to the system rather than substantially altering marginal price incentives for existing recipients.

These reforms occurred in a market characterised by inelastic short-run supply, regulated quality standards, and persistent cost pressures in wages and operating expenses. In such environments, subsidy expansions may be transmitted into observed fees through changes in demand conditions, capacity utilisation, or provider pricing responses, although the under-

lying mechanisms are not directly observed.

Using a quarterly panel of Australian Statistical Area Level 3 (SA3) regions across capital cities from 2018 to 2024, this paper estimates a panel interrupted time series model with unit fixed effects and flexible time trends. The specification allows for both level shifts and changes in fee growth following each reform, capturing dynamic adjustments in observed real childcare fees across regions.

The empirical results are interpreted as policy-associated deviations in fee trajectories relative to underlying trends, rather than fully isolated causal effects. This reflects the presence of strong pre-existing upward trends in childcare fees and the difficulty of fully disentangling policy timing from broader structural and macroeconomic influences in the sector.

Overall, the analysis documents systematic changes in the evolution of real childcare fees around major subsidy reforms and highlights heterogeneity in fee dynamics across policy designs, contributing to understanding of how subsidy structures are reflected in observed pricing outcomes in regulated service markets.

2 Literature Review

A substantial body of literature examines the effects of childcare subsidies on labour supply, childcare utilisation, and household welfare. A central concern in this literature is whether demand-side subsidies translate into lower out-of-pocket costs for households or are partially offset through higher provider prices. Standard public economics theory predicts that subsidy incidence depends on relative elasticities of supply and demand, with the possibility of partial pass-through to prices in markets where supply is relatively inelastic in the short run (Fullerton and Metcalf, 2002).

Empirical evidence from childcare markets suggests that subsidy pass-through is often incomplete. International studies provide consistent evidence that childcare subsidies increase

utilisation but may also be associated with increases in provider fees, particularly in markets characterised by limited competition or capacity constraints (Baker, Gruber and Milligan, 2008; Herbst and Tekin, 2014). This suggests that the intended welfare gains from subsidies may be partially attenuated by equilibrium price adjustments.

In the Australian context, early policy analysis by the Productivity Commission (2014) raised concerns that childcare subsidies, including earlier forms such as the Child Care Benefit and Child Care Rebate, were associated with sustained growth in fees. However, this evidence was largely descriptive and did not isolate causal effects of specific reforms. More recent institutional and policy reviews strengthen this concern. The Productivity Commission (2023) and the Australian Competition and Consumer Commission (2023) document continued increases in childcare fees alongside successive subsidy expansions, highlighting that affordability gains may be partially offset by provider pricing responses in markets characterised by local market power, high regulatory constraints, and limited short-run supply responsiveness. Similarly, the Grattan Institute (2022) argues that structural supply constraints in the childcare sector limit the extent to which subsidies translate into lower effective prices for households.

Within the Australian literature, the dominant focus has been on how childcare subsidies affect childcare utilisation and maternal labour force participation rather than on price responses (Breunig et al., 2011; Doan 2026).

Despite these insights, the existing literature remains largely descriptive or focused on labour market outcomes, with limited evidence on how different subsidy designs are associated with changes in childcare fee dynamics across regions and over time. In particular, there is little evidence distinguishing between subsidy reforms that increase generosity within existing eligibility groups and those that expand eligibility more broadly, or on how these reforms affect both the level and dynamic path of childcare fees across heterogeneous geographic markets.

This paper contributes to the literature by estimating the dynamic effects of two major

childcare subsidy reforms on real childcare fees using a panel interrupted time series framework across Australian capital city SA3 regions. By distinguishing between level and trend effects, and comparing alternative policy designs, the analysis provides new evidence on the extent, timing, and heterogeneity of subsidy pass-through into provider pricing behaviour in a highly regulated service market.

3 Theoretical Framework

3.1 Policy, Prices, and Market Behaviour

This study evaluates how childcare subsidy policies affect provider pricing behaviour. In the Australian childcare market, providers set fees subject to demand conditions, cost structures, and institutional features such as teacher child ratios and reimbursement rules. Government subsidies alter the effective price faced by households and can therefore shift demand for childcare services.

Standard economic theory predicts that when subsidies reduce out-of-pocket costs, demand increases. In the presence of supply constraints, imperfect competition, or capacity limitations this may be associated with upward pressure on observed fees. Consequently, policy changes may affect both the level of fees and their evolution over Time.

3.2 Real vs Nominal Prices

The relevant outcome for policy evaluation is the real price of childcare services. Nominal prices reflect both underlying pricing behaviour and general inflation, whereas real prices isolate changes in economic value. Therefore, the analysis focuses on inflation-adjusted (real) fees to identify policy-induced changes in provider pricing, net of macroeconomic price movements.

3.3 Dynamic Adjustment and Structural Breaks

Policy interventions are modelled as structural breaks in the data-generating process. Each policy may generate two types of effects:

- **Level effects**: immediate shifts in the average fee following policy implementation.
- **Trend effects**: changes in the growth rate of fees after the policy is introduced.

These dynamics reflect gradual adjustment processes, such as capacity expansion, delayed price responses, or changes in competitive behaviour. The inclusion of post-intervention trend variables allows the model to capture these evolving effects.

3.4 Heterogeneity Across Geographic Units

Childcare markets are spatially heterogeneous. Differences in population density, labour markets, income levels, and service availability imply that pricing responses may vary across regions. To account for this, the model includes unit fixed effects that capture Time-invariant differences across geographic areas.

In extended specifications, unit-specific trends are included to allow for differential growth trajectories across regions. This relaxes the assumption that all areas follow a common pre-policy trend.

3.5 Identification Strategy

The causal interpretation of the estimated policy effects relies on the assumption that, conditional on unit fixed effects and time controls, there are no unobserved factors that simultaneously influence both policy exposure and fee outcomes.

Formally, the identifying assumption is:

$$\mathbb{E}[\varepsilon_{it} \mid \text{HCCS}_{it}, \alpha_i, \lambda_t] = 0 \tag{1}$$

$$\mathbb{E}[\varepsilon_{it} \mid \text{CCCB}_{it}, \alpha_i, \lambda_t] = 0 \tag{2}$$

This assumption is strong in the context of persistent upward trends in childcare fees, and results should be interpreted as policy-associated deviations from underlying trends rather than strictly causal effects

4 Data

This study uses a quarterly panel dataset of Australian Statistical Area Level 3 (SA3) regions from the December quarter of 2018 to the December quarter of 2024. The dataset combines childcare fee data from the Australian Department of Education with Consumer Price Index (CPI) data from the Australian Bureau of Statistics (ABS). To ensure consistent inflation adjustment and comparability across regions, the analysis is restricted to SA3 regions located within major Australian capital cities.

4.1 Childcare Fee Data

Childcare fee data are obtained from the Australian Department of Education Quarterly Reports. The analysis focuses on the variable *Mean fee per hour*, which measures the average gross hourly childcare fee charged per child in each SA3 region. Since the Department of Education reports gross fees prior to subsidy reimbursement, the data capture provider pricing behaviour rather than household out-of-pocket costs.

While the raw dataset contains additional information on service counts, fee growth, and the proportion of services charging above the CCS fee cap, these variables are not used in the baseline specification.

The dependent variable in the analysis is the average hourly childcare fee expressed in real 2024 Australian dollars.

4.2 Inflation Data and Real Fees

Inflation data are sourced from the ABS *Consumer Price Index, Australia* publication. Real childcare fees are constructed using capital city CPI measures and expressed in constant 2024 dollars:

$$real_fee_{it} = nominal_fee_{it} \times \frac{CPI_{2024}}{CPI_t}$$

where CPI_t is the CPI in quarter t and CPI_{2024} is the December quarter 2024 CPI index level.

Using capital city CPI measures allows the analysis to more accurately adjust for regional differences in inflation across metropolitan childcare markets.

4.3 Geographic Structure

The analysis is conducted at the SA3 level under the Australian Statistical Geography Standard (ASGS). SA3 regions are designed to represent functional economic or urban subregions and provide a suitable level of geographic detail for analysing variation in childcare fees across metropolitan areas.

Restricting the sample to capital city SA3 regions improves comparability between areas and ensures that each observation can be matched consistently to an appropriate CPI series.

4.4 Policy Variables

The analysis evaluates two childcare subsidy reforms: the Higher Child Care Subsidy (HCCS) and the Cheaper Child Care Bill (CCCB).

For each policy, a treatment indicator is constructed that equals one following policy implementation and zero otherwise. Additional post-treatment trend variables are included to capture changes in fee growth after each reform.

4.5 Data Limitations

The June and September quarters of 2020 are excluded from the sample due to disruptions associated with COVID-19 policy responses and temporary changes in childcare arrangements during the pandemic.

In addition, ASGS geographic boundaries changed between the 2016 and 2021 classifications, which may introduce minor inconsistencies over time. Finally, the analysis focuses on gross provider fees rather than net household childcare costs after subsidies.

5 Methodology

This study examines the impact of two policy interventions on fees using a panel interrupted time series framework. The analysis exploits variation across geographic units over time and estimates models using a fixed effects estimator implemented in the `plm` package.

Let i index geographic units and t index Time. The baseline specification is:

$$\text{real_fee}_{it} = \beta_1 \text{HCCS}_{it} + \beta_2 \text{CCCB}_{it} + \gamma t + \delta_1 \text{TimePostHCCS}_{it} + \delta_2 \text{TimePostCCCB}_{it} + \alpha_i + \varepsilon_{it} \tag{3}$$

where α_i captures time-invariant unit-specific effects and ε_{it} is the idiosyncratic error term. The variables HCCS_{it} and CCCB_{it} are binary indicators equal to one after the implementation of each respective policy. The variables TimePostHCCS_{it} and TimePostCCCB_{it} capture post-intervention slope changes and are defined as:

$$\text{TimePost}_{kit} = \max(0, t - T_k), \quad k \in \{HCCS, CCCB\} \tag{4}$$

where T_k denotes the implementation time of policy k.

Table 1: Regression Results: Real Fee

Variable	Coefficient
HCCS	0.2789***
	(0.0088)
CCCB	0.0168*
	(0.0092)
Time	0.1581***
	(0.0020)
TimePostHCCS	0.2132***
	(0.0038)
TimePostCCCB	-0.0418***
	(0.0046)

Standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

$R^2 = 0.98868$, $n = 151$, $T = 23$, $N = 3473$

Unit fixed effects included. Standard errors clustered at the SA3 level.

6 Results

6.1 Baseline Estimates

Table 1 presents the baseline interrupted time series estimates of the effects of the Higher Child Care Subsidy (HCCS) and the Cheaper Child Care Bill (CCCB) on real hourly childcare fees.

The results indicate that the introduction of the HCCS was associated with a statistically and economically significant increase in childcare fees. The estimated coefficient is consistent with an increase in observed real fees coincident with the HCCS introduction of approximately \$0.28 per hour in real 2024 dollars following implementation. Relative to average hourly fees over the sample period, this represents a modest but meaningful increase in provider prices.

The post-HCCS trend coefficient is also positive and highly statistically significant, indicating that fee growth accelerated after the introduction of the policy. Real childcare fees increased by an additional \$0.21 per quarter beyond the underlying time trend. This suggests that the effects of the subsidy reform accumulated gradually over time rather than

occurring entirely at implementation.

These findings are consistent with the theoretical expectation that subsidy expansions increase effective demand for childcare services. In markets characterised by supply constraints, rising labour costs, and limited short-run capacity expansion, providers may partially pass increased subsidy benefits through into higher prices.

In contrast, the estimated effects of the CCCB are considerably smaller. While the level effect remains positive and marginally statistically significant, the magnitude is economically modest relative to the HCCS. The negative coefficient on the post-CCCB trend variable further suggests that fee growth slowed following implementation of the policy.

The estimated common time trend remains positive and statistically significant across specifications, indicating that childcare fees continued to rise throughout the sample period even after accounting for policy interventions. This likely reflects broader cost pressures within the childcare sector, including wages, rents, and compliance costs.

6.2 Dynamic Effects

The post-treatment trend coefficients imply that the effects of childcare subsidy reforms accumulated over time rather than occurring solely at implementation. In particular, the positive post-HCCS trend effect indicates a persistent increase in the growth rate of real childcare fees following the policy change.

For a household using 30 hours of childcare per week, the estimated HCCS level effect of \$0.28 per hour corresponds to an immediate increase of approximately \$8.40 per week in real childcare fees. The dynamic effects are substantially larger over time. The estimated post-HCCS trend effect implies an additional increase of approximately \$1.50 per hour after two years relative to the pre-policy trend. For a household using 30 hours of care per week, this corresponds to roughly \$45 in additional weekly childcare costs, or approximately \$2,340 per year in real 2024 dollars.

The pattern is consistent with gradual adjustments in observed fee dynamics following

policy implementation, potentially reflecting delayed market adjustment, evolving demand conditions, or ongoing cost pass-through. The results therefore indicate that the effects of childcare subsidies on provider pricing behaviour may emerge progressively over time rather than immediately following implementation.

In contrast, the negative post-CCCB trend coefficient suggests that the growth rate of childcare fees moderated following the introduction of the CCCB. Although fees continued to increase overall, the rate of growth slowed relative to the post-HCCS period, indicating weaker cumulative price effects under the later reform.

7 Robustness Checks

7.1 Inflation Sensitivity

The baseline specification uses inflation-adjusted childcare fees expressed in constant 2024 dollars. Real fees are constructed by deflating nominal fees using the Consumer Price Index (CPI), such that:

$$\text{real_fee}_{it} = \text{nominal_fee}_{it} \times \frac{\text{CPI}_{2024}}{\text{CPI}_t} \tag{5}$$

where CPI_t denotes the price index in period t, and CPI_{2024} represents the index in the base period (December quarter 2024). This transformation expresses all fees in constant purchasing power terms. To assess sensitivity to inflation adjustment, the model is re-estimated using nominal fees:

$$\text{nominal_fee}_{it} = \beta_1 \text{HCCS}_{it} + \beta_2 \text{CCCB}_{it} + \gamma t + \delta_1 \text{TimePostHCCS}_{it} + \delta_2 \text{TimePostCCCB}_{it} + \alpha_i + \varepsilon_{it} \tag{6}$$

Table 2 compares estimates using real (inflation-adjusted) and nominal childcare fees. The results show that while both specifications identify statistically significant effects, the magnitude of the estimated policy impacts is substantially larger in real terms than in

Table 2: Regression Results: Real Fee vs Nominal and Difference

	Real Fee	Nominal Fee	Difference
HCCS	0.279***	0.033***	0.246
	(0.009)	(0.008)	
CCCB	0.017*	0.074***	-0.057
	(0.009)	(0.010)	
Time	0.158***	0.124***	0.034
	(0.002)	(0.002)	
TimePostHCCS	0.213***	0.074***	0.139
	(0.004)	(0.004)	
TimePostCCCB	-0.042***	0.042***	-0.084
	(0.005)	(0.004)	

Standard errors in parentheses (models only).
Difference = Real Fee - Nominal Fee (SE not shown).
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

nominal terms.

For the HCCS, the estimated level effect is 0.279 in real terms compared to 0.033 in nominal terms, indicating that nominal estimates substantially understate the true economic effect of the policy once general inflation is accounted for. A similar pattern is observed for the post-treatment trend effects, where real specifications show stronger and more persistent increases in fees.

This divergence suggests that a significant portion of observed nominal fee growth is driven by broader inflation rather than policy-induced pricing behaviour. Once adjusted for inflation, the results more clearly reflect changes in provider behaviour in response to subsidy reforms.

Overall, the comparison highlights that real fees are the appropriate outcome measure for identifying subsidy pass-through effects, as nominal estimates confound policy impacts with economy-wide price growth.

7.2 Geographic Heterogeneity

7.2.1 Spatial Correlation

Given the geographic structure of the data, error terms may exhibit both serial correlation within units and cross-sectional dependence across units. While the baseline specification employs standard errors clustered at the geographic unit level, this approach does not account for potential correlation across regions.

To address this, Driscoll–Kraay standard errors are implemented. These standard errors are robust to general forms of spatial and temporal dependence, allowing for cross-sectional correlation that may arise from shared economic conditions or spillovers between neighbouring regions. Comparing inference under clustered and Driscoll–Kraay standard errors provides a test of whether spatial dependence materially affects statistical significance.

7.2.2 Regional Trends

To account for heterogeneous underlying trends across geographic units, the model is augmented with unit-specific linear Time trends. This is implemented through the inclusion of an interaction between unit identifiers and Time (denoted as unit_Time$_{it}$), allowing each region to follow its own deterministic trend:

$$\text{real_fee}_{it} = \beta_1 \text{HCCS}_{it} + \beta_2 \text{CCCB}_{it} + \gamma t + \delta_1 \text{TimePostHCCS}_{it} + \delta_2 \text{TimePostCCCB}_{it} + \theta_i(\text{t}_i) + \alpha_i + \varepsilon_{it} \tag{7}$$

This specification relaxes the assumption of common trends across units and reduces the risk that estimated treatment effects are driven by differential pre-treatment trajectories.

Table 3 reports robustness checks that account for spatial correlation and heterogeneous regional trends. Across all specifications, the estimated effects of the HCCS and CCCB remain remarkably stable in both magnitude and significance, suggesting that the baseline results are not driven by spatial dependence or unobserved regional growth patterns.

When Driscoll–Kraay standard errors are applied to account for cross-sectional depen-

Table 3: Regression Results: Robustness Tests

	Real Fee	Spatial Correlation	Regional Trends
HCCS	0.279***	0.279***	0.279***
	(0.009)	(0.043)	(0.009)
CCCB	0.017*	0.017	0.017*
	(0.009)	(0.072)	(0.009)
Time	0.158***	0.158***	0.152***
	(0.002)	(0.005)	(0.003)
TimePostHCCS	0.213***	0.213***	0.213***
	(0.004)	(0.007)	(0.004)
TimePostCCCB	-0.042***	-0.042*	-0.042***
	(0.005)	(0.022)	(0.005)
Unit_Time			0.00007*
			(0.00004)

Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

dence, the coefficients remain statistically significant, although standard errors increase slightly. This indicates that while childcare fee shocks may be correlated across neighbouring statistical areas, such correlation does not meaningfully affect the estimated policy impacts.

Similarly, the inclusion of unit-specific linear trends does not materially change the estimated coefficients. The persistence of the HCCS effect under this specification suggests that the observed increase in real childcare fees is not driven by pre-existing differential growth trajectories across SA3 regions. Instead, fees exhibit a consistent upward response to the policy shock even after allowing each region to follow its own underlying trend.

Ultimately, the robustness of the estimates across both spatial and regional trend adjustments supports the interpretation that the rise in real childcare fees reflects a broad-based and systematic increase across statistical areas, rather than being an artefact of spatial spillovers or heterogeneous regional growth dynamics.

7.3 Placebo Treatments

Placebo tests are conducted by assigning pseudo-intervention dates in the pre-treatment period. The model is re-estimated using these fictitious treatment indicators to assess whether

similar effects arise in the absence of actual policy changes.

Table 4: Regression Results by Model Specification

	Real Fee	2019 Q3	2020 Q1	2021 Q1
Subsidy	0.279***	-0.437***	-0.416***	-0.218***
	(0.009)	(0.014)	(0.012)	(0.011)
Time	0.158***	0.111***	0.143***	0.141***
	(0.002)	(0.006)	(0.003)	(0.002)
TimePostSubsidy	0.213***	0.189***	0.176***	0.195***
	(0.004)	(0.006)	(0.004)	(0.003)

Table 4 reports placebo tests using pseudo-intervention dates in the pre-treatment period. These tests assess whether similar effects can be detected even when no actual policy change occurs.

The results show that several placebo coefficients are statistically significant, which limits their usefulness as a strong robustness check in isolation. This reflects the presence of strong underlying time trends in childcare fees, which can generate statistically significant estimates even when treatment is randomly assigned in time.

However, the magnitudes of the placebo effects are generally smaller than the baseline estimates, and in some cases the estimated treatment coefficients even change sign. For example, in certain placebo specifications the treatment effect becomes negative, implying a reduction in fees following a supposed subsidy increase. This is inconsistent with economic intuition, as a generous subsidy reform would be expected to directly increase provider fees in the short run.

These sign reversals reinforce that the statistically significant placebo results are likely driven by pre-existing trends rather than genuine treatment effects. In contrast, the baseline estimates consistently imply increases in real childcare fees following the HCCS, aligning with theoretical expectations about subsidy-driven demand pressures and partial pass-through in a constrained market.

Overall, while the presence of statistically significant placebo coefficients highlights the importance of strong underlying time dynamics, the implausible signs and smaller magni-

tudes relative to the true treatment effects support the validity of the main results.

8 Discussion

The results suggest that childcare subsidy reforms are associated with increases in real childcare fees, consistent with patterns that could reflect partial pass-through mechanisms in a constrained service market. The estimated effects following the HCCS align with the theoretical prediction that subsidies which reduce out-of-pocket costs increase effective demand for childcare services, placing upward pressure on prices when supply is relatively inelastic.

In the Australian childcare sector, supply is constrained in the short run due to staffing requirements, regulatory standards, and capacity limits within centres. When subsidies expand affordability, households increase demand for childcare hours, and providers may face conditions consistent with upward pressure on fees in response to increased demand. The observed patterns are consistent with demand-side mechanisms, although alternative explanations such as concurrent cost pressures cannot be ruled out.

The dynamic effects further suggest that adjustment occurs gradually. Rather than an immediate one-off change, fees continue to rise after implementation, indicating that providers respond over time as occupancy rates increase and contracts are renegotiated. This pattern is consistent with slow-moving market adjustment rather than instantaneous price resetting.

At the household level, the Higher Child Care Subsidy (HCCS) operates through a more intensive restructuring of subsidy generosity within the existing payment system. Families continue to pay childcare fees upfront, with the government subsidising a larger proportion of the hourly rate directly to providers on behalf of eligible households. A key design feature of the HCCS is the introduction of sibling (multi-child) clauses, which increase the subsidy rate for families with more than one child in care. This creates a non-linear pricing structure in which the marginal cost of an additional child in care is lower for larger families, effectively

strengthening incentives for higher-intensity childcare use. As a result, the policy does not simply reduce average costs, but alters the marginal price schedule faced by households, which is particularly relevant for decisions about additional days or hours of care.

In contrast, the Cheaper Child Care Bill (CCCB) expands subsidy eligibility through two main channels: increasing the maximum subsidy rate and extending eligibility to higher-income households. This represents a broadening of access to the subsidy system rather than a targeted change in marginal incentives for existing users. For households, the CCCB reduces effective childcare costs for a wider group, but the change is less concentrated among high-intensity users who already receive substantial support under the pre-existing system.

Importantly, the behavioural implications differ across the two policies. The HCCS primarily affects the intensive margin of childcare use by lowering the marginal cost of additional hours, particularly for households with multiple children through the sibling discount structure. This directly strengthens incentives for existing users to increase childcare consumption. In contrast, the CCCB operates more through the extensive margin by expanding eligibility and increasing subsidy coverage for higher-income households, many of whom were either previously unsubsidised or only lightly subsidised. As a result, it has a weaker effect on marginal usage decisions among existing childcare users. This difference in margin of impact provides a plausible explanation for the stronger and more persistent estimated effects associated with the HCCS in the empirical results.

These mechanisms provide a plausible explanation for why the CCCB generates smaller and less persistent effects in the estimates. The results suggest that not all subsidy reforms translate into equivalent pricing responses; the way a policy is implemented and perceived by households can influence the extent to which it affects market behaviour.

The robustness checks support the interpretation that the HCCS effect is not driven by model specification or spatial dependence, while the placebo results highlight the presence of strong underlying time trends in childcare fees. These trends reinforce the importance of distinguishing policy-induced effects from broader price dynamics when evaluating subsidy

reforms in service-based markets.

9 Conclusion

This paper examines how real childcare fees in Australian capital city regions evolve around two major subsidy reforms: the Higher Child Care Subsidy (HCCS) and the Cheaper Child Care Bill (CCCB). Using a quarterly SA3-level panel from 2018 to 2024, the analysis estimates a panel interrupted time series model with unit fixed effects and flexible time trends to examine changes in both the level and growth rate of fees.

The results show that observed real childcare fees increase following the introduction of the HCCS, with both immediate level shifts and sustained changes in the post-policy trend. In contrast, the CCCB is associated with smaller and less persistent changes in fee dynamics. Across specifications, these patterns remain stable under alternative treatments of spatial correlation and the inclusion of unit-specific trends.

Importantly, the analysis also highlights the presence of strong underlying upward trends in childcare fees over the sample period. Placebo tests using pseudo-intervention dates indicate that fee dynamics exhibit significant time dependence, which limits the extent to which observed changes can be cleanly separated from broader structural growth in prices. As a result, the estimated coefficients should be interpreted as policy-associated changes in fee trajectories rather than strictly causal effects isolated from all confounding trends.

From an economic perspective, the findings are consistent with a setting in which subsidy expansions coincide with changes in observed pricing dynamics in a constrained service market. However, the results do not distinguish between alternative underlying mechanisms, such as supply-side cost pressures, demand shifts, or partial subsidy pass-through. Instead, they document robust associations between policy timing and changes in the level and growth rate of childcare fees across metropolitan regions.

Overall, the evidence suggests that childcare subsidy reforms are associated with mean-

ingful changes in fee dynamics, but that these changes occur in an environment characterised by strong pre-existing trends and structural market pressures. This highlights the importance of distinguishing policy-related movements from broader inflationary and sector-specific dynamics when evaluating the effects of subsidy reforms in service-based markets.

A Appendix

	SYD	MEL	BRI	ADE	PER
HCCS	0.220***	0.289***	0.289***	0.242***	0.415***
	(0.015)	(0.015)	(0.018)	(0.016)	(0.022)
CCCB	0.035***	-0.022	0.042*	0.051**	-0.029
	(0.011)	(0.014)	(0.024)	(0.025)	(0.033)
Time	0.151***	0.149***	0.186***	0.153***	0.165***
	(0.003)	(0.004)	(0.006)	(0.004)	(0.004)
TimePostHCCS	0.225***	0.222***	0.216***	0.205***	0.193***
	(0.008)	(0.007)	(0.009)	(0.010)	(0.009)
TimePostCCCB	-0.051***	-0.041***	-0.078***	-0.035***	0.006
	(0.008)	(0.008)	(0.012)	(0.012)	(0.009)

Table 5: Interrupted Time Series Estimates of Childcare Fees by City

Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.